

 quarter = 4 equal size pieces.

1 litre = 1.06 ^{cuarto} quarts

4 quarts = 1 gallon

rounded to 108th how many litres per 1 gallon.

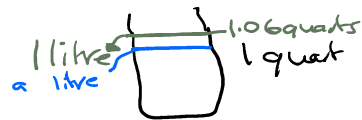
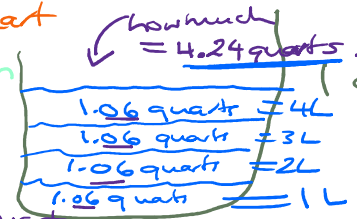
0.943 litre = 1 quart

3.77 litre = gallon

4 quarts = 1 gallon

"a" litres = 1 quart
1 litre = 1.06 quarts

$$a = \frac{1}{1.06} = 0.943$$



$$\begin{aligned} a \text{ litre} &= 1 \text{ quart} \\ 1 \text{ litre} &= 1.06 \text{ quarts} \\ a \text{ litres} &= \frac{1}{1.06} \\ &= 0.943 \text{ litres} \end{aligned}$$

in 1 quart
4 quarts = 1 gallon

$$\begin{aligned} 4 \text{ litres} &= 4 \times 0.943 \\ &= 3.77 \text{ litres (gallon)} \end{aligned}$$

To make a drink

6-8 fl oz



6-8 fl oz per pouch mix

3 cups of water to 4 pouches of mix

1 cup = 8 fl oz
1 cup = 8 fl oz
1 cup = 8 fl oz
24 fl oz
4 pouches of mix

how much water do I need for 4 pouches?

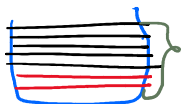
1 cup = 8 fl oz

fl = fluid
= liquid
oz = measure
= volume.

Unit	Sample Use	Comparing Units
fluid ounce (fl oz)	 1 fl oz	1 fl oz = $\frac{1}{8}$ c
cup (c)	 1 cup 8 fl oz	1 c = 8 fl oz
pint (pt)	 1 pt	1 pt = 2 c
quart (qt)	 1 qt	1 qt = 2 pt
gallon (gal)	 1 gal	1 gal = 4 qt

$$\begin{aligned} 24 \text{ fl oz of water} &= \frac{6 \text{ fl oz}}{\text{pouch of mix}} \\ 4 \text{ pouches of mix} & \end{aligned}$$

1 gallon = 3.77



3 x amount

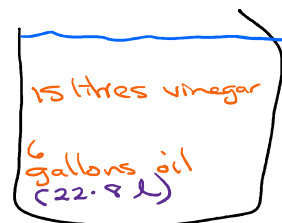
oil : vinegar
2 gallons = 5 litres
6 gallons : 15 litres
22.8 L : 15 L

$$\begin{aligned} 3.8 \text{ litres} &= 1 \text{ gallon} \\ 22.8 \text{ litres} &= 6 \text{ gallons (oil)} \end{aligned}$$

$$\begin{aligned} \text{Total litres} &= 15 + 22.8 \\ &= 37.8 \text{ litres} \end{aligned}$$

oil : Vinegar

2 gallons = 5 litres



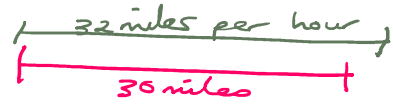
gallons oil
(22.8 L)

oil Vinegar

2 gallons = 5 litres

3x amount
 $\begin{matrix} 7.54 \text{ litre} = 5 \text{ litres} \\ 22.62 = 15 \text{ litres} \end{matrix} \times 3$

drive 32 mph average place is 30 miles away
 How long to get there?



miles hour
 $\begin{matrix} 32 & : & 1 \\ 30 & : & x \end{matrix}$

hours miles
 $\frac{x}{1} : \frac{30}{32}$

$x = \frac{30}{32}$
 $= \frac{15}{16} = 0.9375$
 Round to 1 hour?

$y \text{ mins} = 0.9375 \text{ hours}$
 $60 \text{ mins} = 1 \text{ hour}$

$60 \times \frac{y}{60} = 0.9375 \times 60$
 $y = 0.9375 \times 60$
 $= 56.25 \text{ mins}$

1. \$150 10% off

$150 = 100\%$
 $15 = 10\%$
 $150 - 15 = \$135$

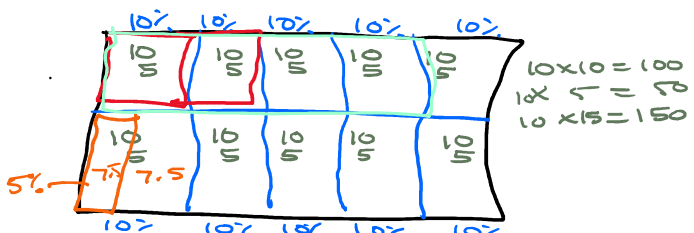


$150 \times 0.9 = 135$
 $(150 \times 0.9 = 135)$

$150 \times 1 = 150$

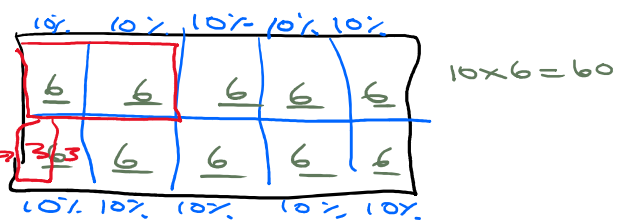
2. What is 10% of \$135

10% of 135 = \$13.50



20% of 150 = 30
 40% of 150 = 60
 5% of 150 = 7.5

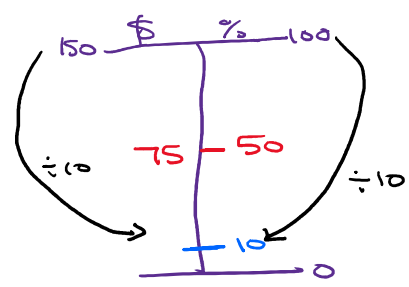
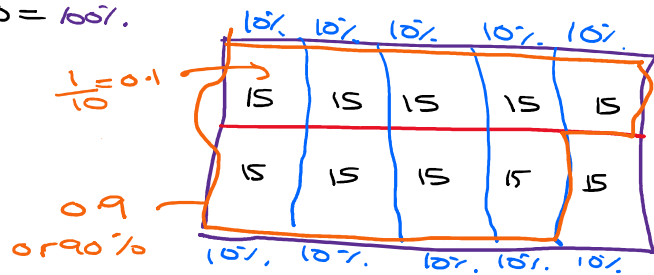
\$60



$90\% = \frac{90}{100}$
 $= \frac{9}{10}$
 $= 0.9$

$100\% - 10\% = 90\%$
 $10\% \text{ of } 60 = 6$
 $20\% \text{ of } 60 = 12$
 $5\% \text{ of } 60 = 3 \rightarrow 26.5$
 $(10\% \text{ off}) = 90\%$
 $60 - 6 = 54$
 $60 - 3 = 57$
 $60 - 54 = 6$

150 = 100%



90% or $\times 0.9 \Rightarrow 150 \times 0.9 = 135$

90% or $\times 0.9 \Rightarrow 150 \times 0.9 = 135$

$$\begin{array}{r} x-3 \\ 2 \overline{) 2x-6} \\ \underline{2x-6} \\ 0 \end{array}$$

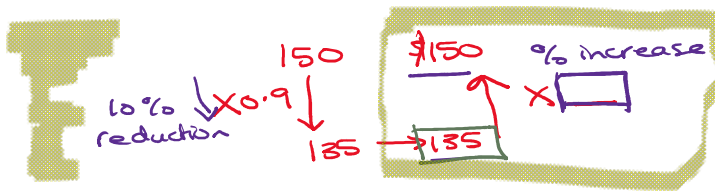
$$\begin{array}{r} x+3 \\ 3 \overline{) 3x+6} \\ \underline{3x+6} \\ 0 \end{array}$$

$$\begin{aligned} 2(x-3) + 4 &= 3(x+2) - 5 \\ 2x - 6 + 4 &= 3x + 6 - 5 \\ 2x - 2 &= 3x + 1 \\ -2 &= x + 1 \\ -3 &= x \\ x &= -3 \end{aligned}$$

Expand (get rid of the brackets)

$$\begin{aligned} 2x - 2 &= 3x + 1 \\ 2x - 2 - 2x &= 3x + 1 - 2x \\ -2 &= x + 1 \\ -2 - 1 &= x + 1 - 1 \\ -3 &= x \\ x &= -3 \end{aligned}$$

Shop sells \$150, 10% off everything
new price = \$135 (150×0.9 or $150 - 15$)
(90%) 100% - 10%



10% of 135 is 13.50

$135 + 13.5 = 148.5$
Not \$150 that I started with!!!
 $\frac{15}{150} = \frac{10}{100} = 10\%$
 $0.9 = 0.90 = 90\%$

\$150 $\rightarrow$ 10% was \$15

so 10% discount = $150 - 15 = 135$

90% of 150 is $150 \times 0.9 = 135$

\$135 what is the % increase to get back to \$150

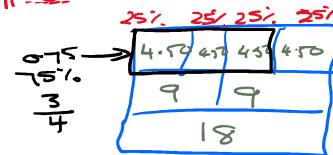
$135 \times \frac{0.112}{100} = 150.12$

1.1111 actually mean

$135 \times \frac{1.1111}{100} = \frac{150}{135}$
 $\frac{x}{100} = \frac{150}{135}$
 $= 1.1111$

$\frac{11.11}{100} = 11.11\%$

Jacket on Sale! 25% off
original \$18 sale price.



calculate 25% & subtract

$18 \times \frac{25}{100} = 4.50$

$18 - 4.50 = 13.50$

multiply to take away 25%

$18 \times 0.75 = 13.50$

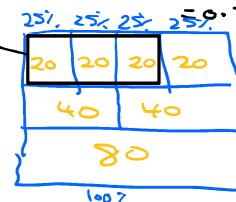
$\frac{75}{100} = \frac{3}{4}$

$\frac{75}{100} = \frac{3}{4}$

$\frac{75}{100} = \frac{3}{4}$

Pants on Sale 25% off
original price ~ \$80
sale price = ?

25% of 80
discount = \$20
sale price = $80 - 20 = 60$
 $80 \times \frac{75}{100} = 80 \times 0.75 = 60$



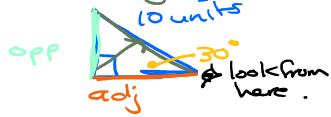
sale price = ?

$$\text{sale price} = \frac{80}{100} = 0.8$$

$$80 \times \frac{75}{100} = 80 \times 0.75 = 60$$

$$\frac{80}{100} = 0.8$$

side opposite the right angle
hypotenuse



opposite = across from
adjacent = next to

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

To find the opposite

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$10 \times \sin(30) = \frac{\text{opp}}{10} \times 10$$

$$10 \sin(30) = \text{opp}$$

$$\text{opp} = 5$$

$$\left. \begin{array}{l} \text{opp} = ? \\ \theta = 30 \\ \text{hyp} = 10 \end{array} \right\}$$

To find the adjacent

$$\cos(\theta) = \frac{\text{adj}}{\text{hyp}}$$

$$\cos(30) = \frac{\text{adj}}{10}$$

$$10 \times \cos(30) = \frac{\text{adj}}{10} \times 10$$

$$\begin{aligned} \text{adj} &= 10 \times \cos(30) \\ &= 5\sqrt{3} \text{ (Exact)} \\ &= 8.66 \text{ (Approx)} \end{aligned}$$

$$\left. \begin{array}{l} \text{adj} = ? \\ \text{hyp} = 10 \\ \theta = 30 \end{array} \right\}$$

You can make only four different cuboids with 16 cubes.

	Dimensions
Cuboid A	1 1 16
Cuboid B	1 2 8
Cuboid C	1 4 4
Cuboid D	2 2 4

(a) Which of the cuboids A and D has the larger surface area?

Tick (✓) the correct answer below.

- ☐ Cuboid A
☐ Cuboid D
☐ Both the same

Explain how you know.

(b) Which cuboid has the largest volume?

Tick (✓) the correct answer below.

- ☐ Cuboid A
☐ Cuboid B
☐ Cuboid C
☐ Cuboid D
☐ All the same

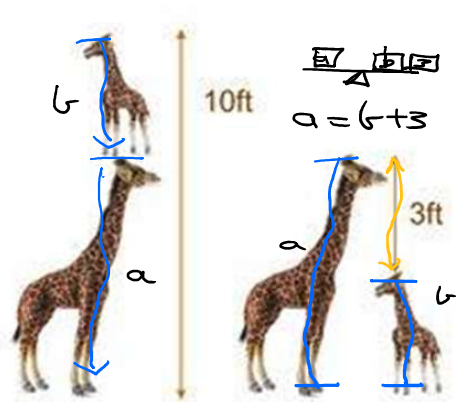
(c) How many of cuboid D make a cube of dimensions 4 x 4 x 4?

(d) You can make only six different cuboids with 24 cubes.

Complete the table to show the dimensions.

Two have been done for you.

	Dimensions
Cuboid E	1 1 24
Cuboid F	1 2 12
Cuboid G	
Cuboid H	
Cuboid I	



find height of each giraffe?

$a = \text{height of the adult giraffe}$
 $b = \text{height of the baby giraffe}$

$$a = b + 3$$

$$a + b = 10$$

$$a - b = 3$$

$$a + a = 2a$$

$$a \times a = a^2$$



$$a + b = 10 \quad (1)$$

$$a - b = 3 \quad (2)$$



$$2a = 13$$

$$a = \frac{13}{2}$$

$$a = 6.5$$

$$\frac{2a}{2} = \frac{13}{2} \quad (\div 2)$$

$$a = 6.5$$

31/1/25

$$a = 6.5$$

$$6.5 + b = 10$$

$$b = 10 - 6.5$$

$$b = 3.5$$

$$6.5 - b = 3$$

$$6.5 - 3.5 = 3$$

$$3 = 3$$



$$2x + 4y = 25$$

$$3x + 3y = 72$$

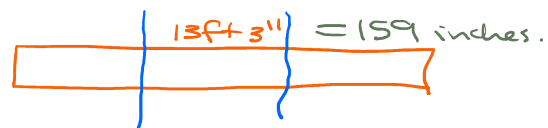


cat

$$6x + 12y = 75$$

$$6x + 6y = 144$$

If a 13 ft board is cut into 3 equal pieces.
 How long is each piece?



Estimate/guess

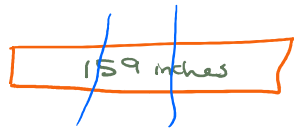
Answer is definitely more than $\frac{4 \text{ ft}}{3 \times 4 = 12}$
 less than $\frac{5 \text{ ft}}{3 \times 5 = 15}$.

13 ft 3 inches $\Rightarrow$ all inches.

1 foot = 12 inches.

$$13 \times 12 \text{ inches} + 3 \text{ inches} = 156 + 3$$

$$= 159 \text{ inches}$$



$$\frac{53}{3 \overline{) 159}}$$

so each pieces is 53 inches

$$53 \text{ inches} = 4 \text{ ft} + 5 \text{ inches.}$$

$$4 \times 12 + 5$$

$$48 + 5$$

Learning is not event, it is a journey

Spain $\Rightarrow$ wonderful maths museum

Stone soup

bean shoot
(mung bean) $\rightarrow$ 1 week

oak tree $\rightarrow$ more than your
acorn lifetime



Patterns aka Algebra

Draw the next pattern in the sequence

Pattern Number (p)

Number of matches in each pattern

Rule $m =$

Number of matches in pattern 10

Pattern 1: $3 \times 1 + 1 = 4$

Pattern 2: $3 \times 2 + 1 = 7$

Pattern 3: $3 \times 3 + 1 = 10$

Pattern 4: $3 \times 4 + 1 = 13$

Matchsticks (m)

Pattern 5: $\text{matchsticks} = 3 \times \text{pattern} + 1$

1 4 7 10 13 16

Draw the next pattern in the sequence

Pattern Number (p)

1 2 3 4 5

Pattern 1: $3 \times 1 + 2 = 5$

Pattern 2: $3 \times 2 + 2 = 8$

Pattern 3: $3 \times 3 + 2 = 11$

Pattern 4: $3 \times 4 + 2 = 14$

Matchsticks (m)

Pattern 5: $\text{matchsticks} = 3 \times \text{pattern} + 2$

Draw the next pattern in the sequence

Pattern Number (p)

1 2 3 4 5

Pattern 1: $3 \times 1 + 1 = 4$

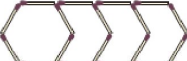
Pattern 2: $3 \times 2 + 1 = 7$

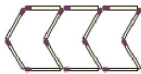
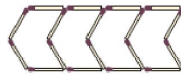
Pattern 3: $3 \times 3 + 1 = 10$

Pattern 4: $3 \times 4 + 1 = 13$

Matchsticks (m)

Pattern 5: $\text{matchsticks} = 3 \times \text{pattern} + 1$

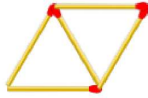
Patterns		Number of matches in each pattern					Rule $m =$	Number of matches in pattern 10
Draw the next pattern in the sequence		Pattern Number (p)						
    Pattern 1 Pattern 2 Pattern 3 Pattern 4 Matchsticks (m)								
Pattern 5								
Draw the next pattern in the sequence		Pattern Number (p)						
    Pattern 1 Pattern 2 Pattern 3 Pattern 4 Matchsticks (m)								
Pattern 5								

		Number of matches in each pattern					Rule m =	Number of matches in pattern 10	
Draw the next pattern in the sequence		Pattern Number (p)							
 Pattern 1		 Pattern 2		 Pattern 3		 Pattern 4			
Pattern 5		Matchsticks (m)							
		1	2	3	4	5			

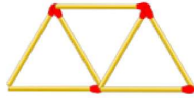
Question 1:



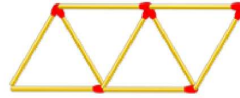
Pattern 1



Pattern 2



Pattern 3



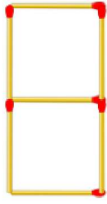
Pattern 4

Pattern Number:	1	2	3	4	...	n	...	100	
Number of Matchsticks:									87

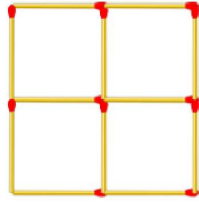
Rule in words: number of matchsticks = _____

Rule in symbols: $m =$ _____

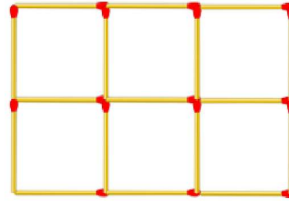
Question 2:



Pattern 1



Pattern 2



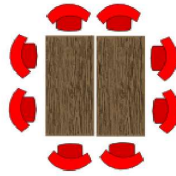
Pattern 3

Pattern Number:	1	2	3	4	...	n	...	114	
Number of Matchsticks:									122

Rule in words: number of matchsticks = _____

Rule in symbols: $m =$ _____

Question 3:



Number of Tables:	1	2	3	4	...	n	...	138
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 4:

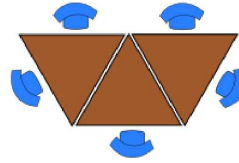
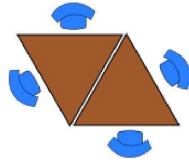


Number of Tables:	1	2	3	4	...	n	...	90
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 5:

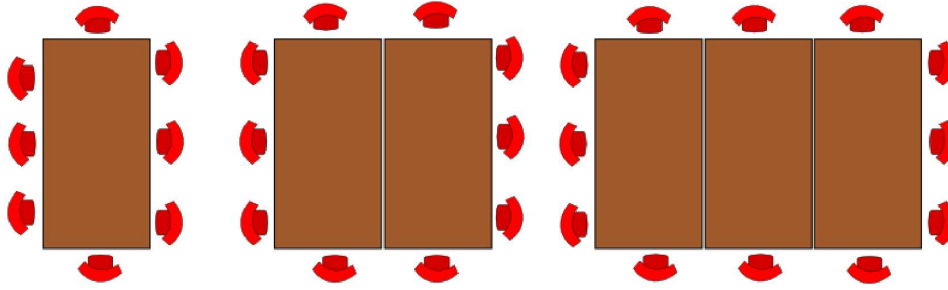


Number of Tables:	1	2	3	4	...	n	...	102
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 6:

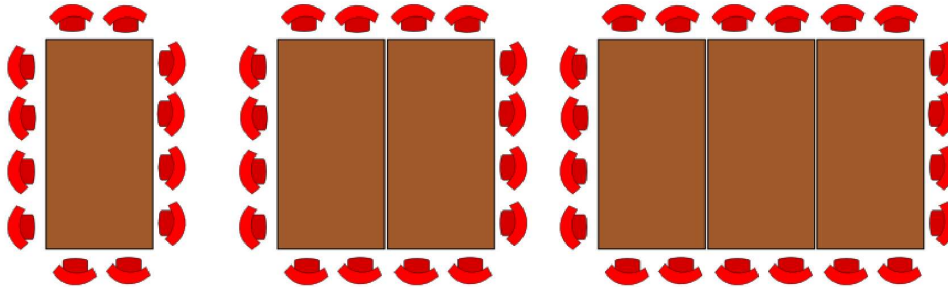


Number of Tables:	1	2	3	4	...	n	...	112
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 7:



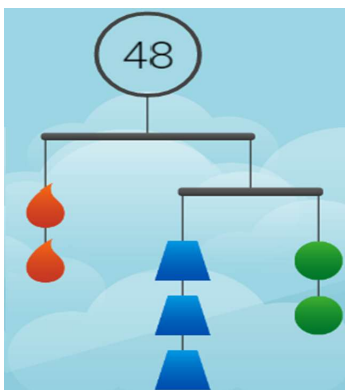
Number of Tables:	1	2	3	4	...	n	...	150	
Number of Chairs:									

In the space below, create your own (random!) pattern and ask another student to work out the rule for the pattern:



- A full jar of honey weighs 1000 grams.
- With half of the honey, the total weight is 600 grams.

What is the weight of the empty jar in grams?





walk at 7 km/h how long does

$$\star \text{ speed} = \frac{\text{distance}}{\text{time}} \quad \triangle \text{ time} = \frac{\text{distance}}{\text{speed}}$$

km/h? distance = speed $\times$ time.
 km per hour? 1 Km 40 mins speed

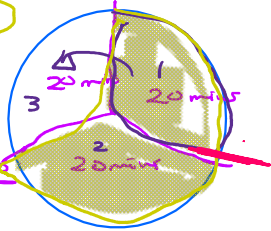
$$1 \text{ km} = 1000 \text{ m}$$

$$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{1 \text{ km}}{40 \text{ min}} = \frac{1000 \text{ m}}{40 \text{ min}} = 25 \text{ m/min}$$

$$\frac{1}{2} = 20 \text{ mins.}$$

$$\text{km/hr} \quad 1 \text{ km} \quad 40 \text{ mins} = \frac{40}{60} = \frac{2}{3} \text{ hr} \quad 60 \text{ mins} = 1 \text{ hour}$$

yellow = 1
 $\frac{1}{2} = \frac{1}{2}$ whole circle
 $\frac{2}{3}$

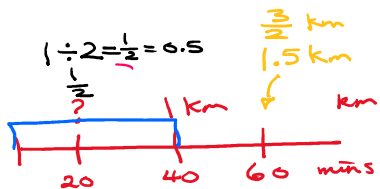


$$\text{speed} = \frac{1 \text{ km}}{\frac{2}{3} \text{ hr}}$$

$$1 \div \frac{2}{3} = 1 \times \frac{3}{2} = \frac{3}{2} \frac{\text{km}}{\text{hr}} = 1.5 \text{ km/hr.}$$

$$6 \text{ km} \Rightarrow \text{takes } 2 \text{ hrs} = 3 \text{ km/hr}$$

$$\frac{3 \text{ km} = 1 \text{ hr}}{3 \text{ km} = 1 \text{ hr}} \Rightarrow \frac{6 \text{ km}}{2 \text{ hrs}} = 3 \text{ km/hr}$$



how far did I run in 20 mins

$$\frac{40 \text{ mins}}{60 \text{ mins/hr}} = \frac{2}{3} \text{ hr}$$

$$1 \text{ km} = 40 \text{ mins.}$$



$$30 \text{ mins} = \frac{30}{60} = \frac{1}{2} \text{ hr} = 0.5 \text{ hr}$$

$$180 \text{ mins} = \frac{180}{60} = 3 \text{ hours}$$

$$60 \text{ } 60 \text{ } 60 = 3 \text{ hrs}$$

$$200 \text{ mins} = \text{---} \text{ hours.}$$

$$\frac{200}{60} = 3.3 \text{ hr} = 3 \text{ hr } 20 \text{ mins}$$

$$\begin{array}{r} \text{length} \\ \hline 100 \end{array} \quad \begin{array}{r} \text{time} \\ \hline 60 \end{array}$$

$$\frac{180}{60} = 3 \text{ hr } 20 \text{ mins} = \frac{1}{3} \text{ hr} = 0.33$$

$$60 \text{ seconds} = 1 \text{ min}$$

$$150 \text{ seconds} = 150 \times \frac{1}{60} = \frac{150}{60} = 2.5 \text{ mins}$$

$\frac{150}{60} = \frac{5 \times 30}{2 \times 30} = \frac{5}{2} = 2\frac{1}{2} = 2.5$

$30 \text{ seconds} = 0.5 \text{ min}$



$\begin{array}{r} 4 \text{ lb } 10 \text{ oz} \\ 3 \text{ lb } 4 \text{ oz} \\ \hline 7 \text{ lb } 14 \text{ oz} \end{array}$	$\begin{array}{r} 4 \text{ lb } 10 \text{ oz} \\ 3 \text{ lb } 7 \text{ oz} \\ \hline 7 \text{ lb } 17 \text{ oz} \\ = 8 \text{ lb } 1 \text{ oz} \end{array}$	$\begin{array}{l} 1 \text{ lb } = ? \text{ oz} \\ 16 \text{ oz} \\ 16 \text{ oz} = 1 \text{ lb} \\ 17 \text{ oz} = 16 \text{ oz} + 1 \text{ oz} \\ = 1 \text{ lb} + 1 \text{ oz} \end{array}$
--	--	---